Neural emulation of SPEX single-element CIE spectra: architecture, training strategy, and evaluation

Technical report on design decisions, including discarded approaches

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Abstract

We document the design of a neural emulator for velocity-broadened, single-element collisional ionisation equilibrium (CIE) X-ray spectra computed with SPEX [Kaastra et al., 1996], developed on the iron ($Z = 26$) model as testbed. The final configuration – a coordinate-based "operator" network with Fourier-feature energy encoding, FiLM temperature conditioning, a dedicated line-amplitude head, error-prioritized example sampling, Polyak weight averaging, and a long training schedule – reaches a mean relative flux error of 0.13% on held-out SPEX spectra (99.2% of test spectra below 1%), at 21.5 ms per spectrum. We lay out each architecture and training decision in the order it was made, with the quantitative evidence behind it, and we give equal space to the approaches that were tried and discarded: a Sobolev derivative-matching loss, PCA+GP emulation, a monolithic (T, v) emulator of broadened spectra, enlarging the line head, synthetic-data augmentation on dense grids, and unweighted per-channel instrument metrics. We further describe validation at the level of real instrument responses (XRISM/Resolve, Chandra ACIS and HETG), the physics argument for per-element velocity broadening, an active-learning scheme for placing new SPEX evaluations, and a feasibility analysis for full 30-element inference within a 200 ms budget.

1 Problem setting and scope

The SPEX CIE model represents an optically thin plasma as a sum of per-element spectra: the observed model spectrum is a linear combination $\sum_i A_i f_i(E; T)$ of single-element spectra f_i weighted by abundances A_i, subsequently velocity-broadened, redshifted, and folded through an instrument response. Fitting such models with microcalorimeter-resolution data is limited by the cost of evaluating f_i; we therefore emulate each element's spectrum with a neural network. This report covers the single-element ($Z = 26$, iron) testbed: data, architecture, training, broadening, and evaluation, in the order decisions were made. Scaling to all 30 elements uses an automated sweep (`scripts/run_all_elements.py`) with the recipe fixed by the experiments below; the sweep and a failure mode it exposed on low-Z elements are in Section 7.3.

Data. 14,246 SPEX spectra on an adaptive energy grid of 24,212 bins over 0.1–12 keV, with temperatures $T = 0.50$–19.94 keV; split 81/9/10 into train/validation/test (11,539 / 1,282 / 1,425 spectra) with a fixed seed. Targets are $\log_{10}$ flux densities; bins with $\log_{10} f < -10$ ("the floor") are treated as empty. Validation and test sets contain only original SPEX spectra in every experiment, including all augmentation studies.

*Drafted with Claude Fable 5; all results produced by the `spexai` codebase, branch `neural_operator`.

Headline metric. For spectrum s, predicted log-density p_{sb} and truth t_{sb},

$$\varepsilon_{sb} = \left|10^{\,d_{sb}} - 1\right|, \qquad d_{sb} = \mathrm{clip}\big(p_{sb} - \max(t_{sb},\,-10),\,-4,\,4\big), \tag{1}$$

which equals the relative error in linear flux, and

$$\mathrm{MRE}_s = \frac{1}{|V_s|}\sum_{b\in V_s} \varepsilon_{sb}, \qquad V_s = \{\,b : t_{sb} > -10\,\}. \tag{2}$$

We report the mean and median of MRE_s over the test set and *yields*, the percentage of spectra with MRE_s below 1% (and 0.1%). Empty bins are scored separately as *floor violations* (fraction of empty bins predicted above the floor, and the worst excess in dex). Line bins (5.9% of the grid; flux exceeding twice a running-median continuum) and continuum bins are also scored separately.

2 Architecture

The emulator is a coordinate-based network in the style of neural operators [Lu et al., 2021, Li et al., 2021] and follows Ricketts et al. [2026]: the spectrum is a continuous function $\log_{10} f(x;\theta)$ of the normalised log-energy coordinate x, conditioned on normalised parameters θ (here $\log_{10} T$). Components, each validated by ablation (Section 2.1):

1. **Fourier-feature encoding of energy** [Tancik et al., 2020]: $x \mapsto [x, \cos 2\pi f_k x, \sin 2\pi f_k x]$ with K log-spaced frequencies $f_k \in [0.25, f_{\max}]$. A coarse-to-fine curriculum weight per channel activates high frequencies progressively during the first 30% of training, preventing early high-frequency overfitting.

2. **FiLM conditioning** [Perez et al., 2018]: a small MLP maps θ to per-layer scale/shift (γ, β) modulating the trunk, rather than concatenating θ to the input.

3. **Trend head**: a linear-in-x term $a(\theta)\,x + b(\theta)$ factoring out the global spectral slope.

4. **Line head**: the line-bin set is precomputed from the training data (union over a temperature-ordered sample; 3,980 lines for Fe, deterministic given the cache). Each line has a learned embedding; a shared MLP maps [embedding, conditioning] to a $\log_{10}$ line flux, which is flux-added (logaddexp) to the trunk output, so the trunk models only continuum and pseudo-continuum. The head stores line energies and native bin widths, so integrated line fluxes can be re-deposited exactly onto arbitrary instrument grids (`deposit()`), a property used heavily in Sections 5 and 8.

5. **Per-bin target normalisation**: the network predicts the z-scored residual of $\log_{10} f$ per training bin; the static line forest lives in the normalisation constants and the trunk models $\mathcal{O}(1)$ deviations.

The training loss is a Huber loss on ε_{sb} (Eq. 1) over P randomly sampled grid points per spectrum, plus a one-sided floor penalty $10^{\mathrm{relu}(p_{sb}+10)}-1$ on empty bins (the model is pushed onto the floor but never penalised for matching it), plus an early log-space MSE stabiliser ramped to zero over the first 20% of steps. Optimiser: AdamW [Loshchilov & Hutter, 2019] with gradient clipping at unit norm and a cosine schedule with linear warmup [Loshchilov & Hutter, 2017].

2.1 Ablation study (what established the components)

An eight-variant ablation at fixed budget (20k steps, $\sim$1M parameters) established, in order of effect size: Fourier features are essential (without them, line MRE is 76%); FiLM is the second-largest factor (yield@1% drops from 88.8 to 60.8 without it); a dedicated line head gives the best line-bin accuracy of the operator family. A multi-resolution grid encoding [Instant-NGP style; Müller et al., 2022] performed like the mid-pack variants but evaluated slower, and was not pursued.

Discarded: Sobolev (derivative-matching) loss. A loss term matching finite-difference slopes between sampled points, natural for smooth reflection spectra, *hurts* here: removing it improved MRE from 0.91% to 0.52%. The CIE spectrum is a forest of single-bin spikes; penalising slope mismatch on subsampled points biases the model against exactly the structure it must fit. All subsequent configurations omit it.

Retained as a deployment option: fixed-grid MLP. The original formulation (an MLP from T directly to all 24,212 bins) is competitive in accuracy (0.53%) and 30× faster (0.2 ms/spectrum) at 15× the parameters. Because CIE line energies never move, the standard argument against fixed grids does not apply; this becomes the distillation target for high-throughput inference (Section 9).

3 Hyperparameter search

A two-stage random search [Bergstra & Bengio, 2012] over 24 configurations (learning rate, batch, points per spectrum, trunk width×depth, activation, K and $f_{\max}$, line-embedding size, trend head, per-bin normalisation, stabiliser weight, curriculum length): stage one at 3k steps, top four re-trained at 20k. Winner `t04`: GELU 384×5 trunk, $K = 512$, $f_{\max} = 4000$, learning rate 3×10^{-3}, batch 128, 2048 points/spectrum, line_dim 16, trend and bin-norm on (1.65M parameters): 0.31% test MRE, 94.7% yield@1%. Validation ranking generalised cleanly to the test set.

Two findings shaped later work. First, *training length dominated every hyperparameter* (0.82% → 0.31% from 3k→20k steps). Second, the largest configuration (`t16`, 3.6M parameters) was competitive at 3k steps but *diverged* at 20k at the shared learning rate of 3×10^{-3}: the search had coupled model size to an unstable learning rate, and the apparent conclusion "larger models do not help" was an artefact (revisited in Section 7).

4 Classical baselines (and a discarded one)

Per-bin interpolation across training temperatures is the natural classical competitor with one physical parameter. Monotone cubic interpolation [PCHIP; Fritsch & Carlson, 1980] and linear interpolation in $(\log T, \log f)$ achieve 0.003% MRE on the full training grid – two orders of magnitude better than any neural variant – and 0.067% with only 300 training spectra. The emulator's justification is therefore never raw single-element accuracy but speed, differentiability, memory (4 MB vs 1.1 GB), and extension to joint multi-parameter spaces where dense tabulation is impossible.

Discarded: PCA + Gaussian process emulation. The Speculator-style approach [Alsing et al., 2020] – PCA on $\log f$, GP regression on component amplitudes – saturates at its truncation error (0.2% with 64 components) and degrades catastrophically at small n_{train} (9–17% MRE). Rejected as both baseline and generator for data augmentation; PCHIP was used instead throughout.

5 Velocity broadening

5.1 Physics: why per-element broadening is correct

Turbulent/bulk velocity broadening acts as a convolution with a kernel K_v in $\ln E$; convolution is linear, so for a shared kernel, broadening per element and summing is *exactly* equivalent to broadening the summed spectrum. Moreover the physically correct total line width is per-ion,

$$\sigma_i^2 = \sigma_{\rm turb}^2 + \frac{kT}{A_i m_p}, \tag{3}$$

with the thermal term scaling as $1/\sqrt{A_i}$ [Rybicki & Lightman, 1979] – the decomposition used in microcalorimeter analyses [Hitomi Collaboration, 2016, 2018] – so a per-element architecture is *more* physical than broadening the total: each element can carry its own thermal width at zero cost. Caveats where the convolution picture itself breaks (multiphase kinematics, resonant scattering) are orthogonal to the per-element choice.

5.2 Benchmark of broadening implementations

Against exact erf-integral broadening of true spectra: FFT convolution on a uniform $\ln E$ grid is accurate to 0.03–0.06% over 100–1000 $\rm km\,s^{-1}$ at 10–30 ms (CPU); the *hybrid* pipeline (emulate unbroadened → FFT-broaden the trunk → add analytic Gaussian lines with exact per-line fluxes) reaches 0.13–0.18%; the legacy sparse-matrix implementation degrades from 0.3% to 2.8% at 1000 $\rm km\,s^{-1}$ with 20% of line flux misplaced. **Recommendation: replace the sparse method with FFT/hybrid.**

Discarded: monolithic (T, v) emulator of broadened spectra (option 2, v1). A two-parameter version of the winning architecture, trained on FFT-broadened targets with the line head removed (rationale: broadening removes delta spikes), reached only 2.6% validation MRE (yield@1% = 0) and 1.1–1.8% end-to-end – 8× worse than the hybrid. The post-mortem identified the causes: (i) velocity is a *known operator*, not a latent parameter – asking global FiLM modulation to express a local convolution discards the strongest available inductive bias; (ii) at $v_{\rm min} = 50\,\rm km\,s^{-1}$ lines are still ∼2 native bins wide, so removing the line head deleted the component handling the sharpest structure while keeping trunk hyperparameters tuned assuming its presence; (iii) uniform point sampling on the 83,728-bin uniform grid starves line cores of supervision; (iv) checkpoint selection never saw the hardest velocities (evaluation at $\{100, 250, 600\}\,\rm km\,s^{-1}$ only); (v) zeroing sub-10^{-9} wings in the targets while penalising above-floor predictions injects label noise at every line wing. A revised version (`train_broadened2`) addresses all five: a *Gaussian line head* with analytic width $\sigma_u = v/c$ and T-only amplitudes (broadening conserves line flux; the factorisation is structural), line-aware point sampling with offsets $\sim \mathcal{N}(0, \sigma(v'))$ at random velocities, validation at $\{50, 100, 250, 600, 1000\}\,\rm km\,s^{-1}$, and wing masking with a one-sided cap. These fixes more than halved the error – v2 reaches 0.46–0.77% end-to-end MRE (Table 1) versus v1's 1.1–1.8%, a ∼2.4× improvement at every velocity, confirming that each diagnosed cause was real. But v2 remains 4–6× worse than the hybrid (0.11–0.13%) and is the slowest method of all (it evaluates the trunk on the full uniform grid plus the line deposits), so the revised option 2 is competent but not competitive.

The conclusion is therefore structural, not a training-quality artifact: even a well-executed monolithic (T, v) emulator cannot beat the hybrid, because the hybrid performs the convolution *exactly* (FFT plus analytic line deposit) while the monolith must *learn* the broadened continuum shape. Broadening belongs *outside* the network; the hybrid on the best unbroadened backbone is the deployment choice, and option 2 is settled as the losing branch – retained here because v2 is the controlled demonstration that the 8× v1 gap was structural.

method	$100\,\mathrm{km\,s^{-1}}$	$300\,\mathrm{km\,s^{-1}}$	$1000\,\mathrm{km\,s^{-1}}$	note
FFT (true input)	0.03%	0.03%	0.06%	accuracy ceiling
hybrid	**0.13%**	**0.11%**	**0.12%**	emulate + analytic broaden
emulator_Tv2 (v2)	0.49%	0.46%	0.77%	monolithic, fixed
sparse (true input)	0.45%	0.30%	2.8%	legacy production
emulator_Tv (v1)	1.17%	1.06%	1.77%	monolithic, original

Table 1: End-to-end broadening MRE vs exact erf broadening. FFT and sparse take the true spectrum as input (isolating the broadening operation); the three emulator methods are end-to-end. The revised monolithic (T, v) emulator (v2) more than halves v1's error but stays 4–6× above the hybrid.

6 Training-data experiments: active learning and augmentation

Motivated by simulator cost for future elements, we tested dynamically generating extra training spectra with the best classical interpolator (PCHIP), with placement informed by model error – an active-learning / adaptive-sampling problem [Liu et al., 2018, Wu et al., 2023, Crombecq et al., 2011, Nygaard et al., 2023, Rogers & Peiris, 2019]. Design: three arms on a 300-point training grid (evenly subsampled in $\log T$), all validated and tested exclusively on original SPEX spectra.

- **baseline**: uniform sampling of the grid;
- **reweight**: examples sampled $\propto$ a running (exponentially averaged) per-spectrum training error, mixed 50/50 with uniform – prioritized sampling in the spirit of Schaul et al. [2016];
- **adaptive**: reweight + gated generation: every 500 steps, new temperatures are drawn with density $\propto$ (local error)^2 [RAD; Wu et al., 2023] from grid intervals whose leave-one-out interpolation error [cf. CV-Voronoi; Liu et al., 2018] is below 1%; PCHIP spectra generated there enter batches at 25% with half loss weight (a low-fidelity teacher in the sense of Kennedy & O'Hagan 2000).

arm	test MRE	yield@1%	synthetic	floor viol.
baseline	0.45%	91.3%	0	0.28%
reweight	0.32%	95.8%	0	0.21%
adaptive	0.30%	96.3%	1152	0.49%
t04_long, full 11.5k grid	*0.31%*	*94.7%*	–	*0.19%*

Table 2: Adaptive-data study ($n_{\mathrm{train}} = 300$). The adaptive arm matches the full-grid model with 2.6% of the simulator budget.

Findings (Table 2): most of the gain is *attention, not data* – prioritized sampling alone closes 85% of the gap at zero cost; the synthetic data adds a small further increment at the price of doubled floor violations (interpolation smears flux into empty bins); and 300 SPEX evaluations plus augmentation match the full-grid model – a $\sim 38\times$ simulator-budget reduction, the practically important result for new elements. The trust gate rejected 23 intervals (clustered at $T = 0.53$–$0.76\,\mathrm{keV}$, leave-one-out errors 2–4%): an explicit shortlist of where only *real* SPEX runs help, closing the loop with simulator-in-the-loop acquisition schemes [Nygaard et al., 2023, Rogers & Peiris, 2019].

Discarded: augmentation on dense grids. On the full grid, interpolation error (0.003%) lies two orders of magnitude below emulator error (0.3%): every synthetic spectrum is, at the network's error resolution, a duplicate of its neighbours. This was argued from the error budget and confirmed by the Tier-1 result below (the model is capacity-limited, not data-limited: train and validation MRE are identical). Augmentation is therefore reserved for sparse-grid regimes ($n_{\mathrm{train}} \lesssim$ a few hundred here).

7 Training strategy: Tier 1 (optimisation) and Tier 2 (capacity)

7.1 Tier 1

Three changes, no architecture change, full training grid: (i) 5× longer schedule (100k steps ≈ 1100 epochs) with a floor on the cosine at 2% of peak learning rate; (ii) Polyak/EMA weight averaging [Polyak & Juditsky, 1992, Izmailov et al., 2018] with decay 0.999 – validation and checkpoints use the averaged weights, which also stabilises the noisy yield-based checkpoint selection; (iii) the prioritized sampling from Section 6. Result: 0.31% → **0.16**% MRE, yield@1% 94.7 → 99.0, yield@0.1% 0 → 43.8, floor violations halved. Training and validation MRE were identical at convergence (0.15%): no overfitting after 1100 epochs, implying a capacity limit, and the best checkpoint was the final step, implying the budget still binds.

Caveat found only by instrument-level tests. Prioritization traded away accuracy on the faint high-energy tails of cool ($T \lesssim 2\,\mathrm{keV}$) spectra – invisible to Eq. 2 (those bins are floor-masked or negligible in the full-band average) but exposed as a 2–5% regression in the Chandra grating bands (Section 8). Mitigated in Tier 2 by lowering the prioritized fraction from 0.5 to 0.4.

7.2 Tier 2

Two capacity hypotheses at equal budget, both with the Tier-1 recipe and a warmup–stable–decay (WSD) schedule [flat learning rate, linear decay over the final 15%; extending a run re-pays only the decay leg; Hu et al., 2024] :

	params	MRE	line/cont	yield@1%	yield@0.1%	ms/spec
tier1 (t04 arch)	1.65M	0.16%	0.21/0.15	99.0	43.8	12.4
big_trunk	3.3M	**0.13%**	0.19/0.12	**99.2**	**52.6**	21.5
line_heavy	1.9M	0.21%	0.23/0.21	98.1	25.8	12.9

Table 3: Tier-2 capacity experiments (100k steps, full grid).

`big_trunk` (512×6, $K = 1024$, learning rate 10^{-3}) wins across the board (Table 3), formally overturning the HPO's size conclusion: it was always the learning rate. Scaling is $\sim 25\%$ MRE reduction per parameter doubling with both runs still budget-clipped, so the next cheap move is the same configuration at 200k steps.

Discarded: enlarging the line head (`line_heavy`). Wider line embeddings (48), a wider shared MLP (256), and Fourier-embedded T-conditioning of line amplitudes – targeting the dominant line-MRE component – performed *worse than Tier 1* (0.21% overall; line MRE 0.23% vs 0.21%). The line-error bottleneck is thus not amplitude-head capacity; the likely residual is the trunk under-resolving inter-line pseudo-continuum. (Minor caveat: this arm ran at the default learning rate 3×10^{-3}; given it regressed rather than plateaued, we did not spend a further run on the learning-rate confound.)

7.3 Multi-element generalisation: floor-dominated spectra

Fixing the recipe on iron and sweeping the low-Z elements ($Z = 2$–6) with the automated per-element sizing exposed a failure mode absent from iron. Two sharply separated tiers emerged (Table 4): helium and carbon reach $\leq 0.04\%$ MRE at 100% yield on both thresholds, while lithium, beryllium, and boron land at 1.3–9.9% MRE with essentially no spectra under 0.1%.

The split is explained entirely by *signal sparsity*, not atomic number. On the shared 24,212-bin grid the fraction of bins ever above the -10 floor is $\sim 97\%$ for He/C (broad continua) but under 1.2% for Li/Be/B – beryllium carries flux in only ~ 6 bins per spectrum, the rest at floor. This is not intrinsic difficulty: classical linear interpolation in T reaches $\sim 10^{-6}$ MRE on all five elements (the trust gate reports 100% of intervals interpolable), so the targets are trivially learnable and the poor emulator numbers are a training pathology.

	Z	signal bins/spec	signal frac.	emul. MRE	interp. MRE
He	2	23,703	97.9%	3.2×10^{-4}	8.8×10^{-7}
C	6	23,444	96.8%	4.0×10^{-4}	8.6×10^{-7}
B	5	292	1.2%	9.9×10^{-2}	1.1×10^{-6}
Li	3	15	0.06%	1.3×10^{-2}	9.1×10^{-7}
Be	4	6	0.03%	5.9×10^{-2}	9.9×10^{-7}

Table 4: Low-Z sweep before the signal-aware fix: emulator accuracy tracks signal density, while classical interpolation is uniformly excellent. Signal bins are per-spectrum bins above the floor (mean over the test set); MRE is the fraction of Eq. 2.

Two compounding causes. (i) Uniform point sampling: each step draws `points`$= 2048$ of the 24,212 bins uniformly, so for beryllium the expected number of non-floor bins seen per step is $2048 \times 0.0003 \approx 0.5$ – the model is supervised on the very bins it is scored on in only half of all steps. (ii) Even when a line bin is drawn, the flat per-point mean in the training loss lets the ~ 2000 trivial floor bins dominate the gradient. Compounding both, the automated size classifier keyed on line-bin *count* and routed Li/Be to a "smooth" preset that switched the line head *off* – disabling the line machinery for precisely the spectra that are almost nothing but lines.

Fix: signal-aware sampling. We replace edge-only sampling with a *signal pool* (union of detected line and recombination-edge bins) and (a) force its members into every batch, and (b) up-weight them per point so they hold a target share `signal_frac` of the loss regardless of how few they are. This is one self-tuning knob that decouples a signal bin's loss *influence* from its sampled *count*, so the same setting holds across beryllium's ~ 6 and boron's ~ 292 signal bins. The size classifier is corrected to route on signal *density*: a floor-dominated spectrum with any lines goes to a new `sparse_lines` preset (line head on, `signal_frac` $= 0.5$), which line-bin count alone cannot distinguish from a genuinely smooth continuum (He). The broad-continuum elements are unaffected – their signal pool spans the band, so the weighting is a near-identity.

8 Instrument-level validation

Native-grid metrics cannot answer whether the emulator is accurate *at the resolution an instrument sees*. We evaluate on (i) resolution-matched grids (bin width = FWHM/2 from published resolutions) and (ii) the actual detector channel grids, folding both truth and prediction through the real redistribution matrices: XRISM Resolve `rsl_Hp_L_2025.rmf` [60000^2, 93M non-zeros; Ishisaki et al., 2025], Chandra ACIS `aciss_aimpt_cy28.rmf` [Garmire et al., 2003], and HETG HEG/MEG grating RMFs [Canizares et al., 2005], parsed from the OGIP format [George et al., 2007]. The instrument metric is Eq. 2 applied to integrated flux per channel with validity mask $F^{\text{true}} > 0$, band-restricted (Resolve: 1.8–12 keV, the closed-gate-valve limit). ARF weighting is deliberately excluded (effective area does not mix channels).

Findings: (i) *median* channel errors (0.03–0.08%) are better than native MRE – rebinning averages independent per-bin errors; (ii) `big_trunk` is the best model on ACIS and both gratings and repairs the Tier-1 grating regression; (iii) at Resolve's 0.5 eV channels, sub-bin line placement – exact-bin deposit vs. flux split across a channel edge – is a real error mode invisible on the native grid; (iv) unweighted per-channel *means* are ill-conditioned: rebinning gives every channel a scrap of flux, so near-empty channels (unbounded relative error on ∼zero truth) and the 4/16 coolest test spectra (negligible in-band flux) dominate, with means swinging by 30× between models whose medians agree.

Discarded: unweighted per-channel means as a headline number. For publication, the flux-weighted error

$$\mathrm{MRE}_s^{\text{flux}} = \frac{\sum_c F_{sc}^{\text{true}}\,\varepsilon_{sc}}{\sum_c F_{sc}^{\text{true}}} = \frac{\sum_c |F_{sc}^{\text{pred}} - F_{sc}^{\text{true}}|}{\sum_c F_{sc}^{\text{true}}} \tag{4}$$

(the misplaced-flux fraction) is immune to empty-channel pathology by construction; implementation is a planned change. Restricting the 1.8 keV Resolve cutoff changed results by $< 2\%$ relative – the pathology lives inside the science band, not below it.

9 Inference feasibility for the 30-element model

Requirement: forward passes through 30 element models, abundance combination, broadening, and response folding in < 200 ms. Measured ingredients: 8.3 ms/spectrum amortised (batch 256, A100, full native grid) for the operator model; Resolve RMF folding is memory-bandwidth bound (93M non-zeros ≈ 0.75 GB of traffic): 37 ms measured on a laptop CPU, ∼0.5–1 ms projected on A100 HBM – reconciling the historical observation that the RMF fold dominated CPU pipelines with its negligible cost GPU-resident. The naive 30-element loop (∼250 ms compute-bound) is brought within budget by (i) evaluating the smooth trunk only at instrument-required resolution (lines deposited analytically; 3–20×), (ii) bf16 and grouped batched matmuls over the 30 weight sets (2–4×), (iii) folding once after summation (linearity), with per-element thermal widths as a single batched FFT. Projected: 30–80 ms per likelihood evaluation. For sampling workloads, distillation into fixed-grid heads (0.2 ms/spectrum; Section 2.1) using the operator models as teachers reduces the full stack to ∼10 ms; the RMF can additionally be truncated at 10^{-5} of row peak (54% of non-zeros, 0.007% flux loss).

10 Related work (2023–2026) and candidate improvements

A survey of recent ML-venue literature identifies candidates in three categories: improvements on deployed strategies, competing alternatives, and additions for performance per unit compute.

Improvements on deployed strategies. (i) Our error-prioritized sampling is a *biased* gradient estimator – the mechanism behind the Tier-1 grating regression. InfoBatch [Qin et al., 2024] performs soft loss-based pruning *with gradient rescaling of survivors*, provably unbiased, saving 25–40% of training cost; it is a small change to our sampler that removes the bias and the `pr_mix` heuristic simultaneously. (ii) Maximum Update Parametrization [Yang & Hu, 2021] transfers learning rates across widths, formally solving the size–learning-rate coupling that misled our HPO (Section 3); it is used for spectral emulators by Różański & Ting [2025], who also report compute-optimal scaling exponents for this problem class. (iii) Schedule-Free AdamW [Defazio et al., 2024] replaces our hand-assembled WSD-plus-EMA combination (no pre-committed budget; averaged-iterate inference built in), and Muon [Jordan et al., 2024] applies orthogonalized updates to weight matrices; recent MLP-regression benchmarks find Schedule-Free, Muon, and AdamW+EMA (our recipe) the three consistent winners, making the other two cheap drop-in trials. (iv) FINER's variable-periodic activations [Liu et al., 2024a] and Fourier-reparameterized training [Shi et al., 2024] address spectral bias adaptively, targeting the fixed-$f_{\max}$ ceiling of our encoding; EVOS [Zhang et al., 2024] formalises coordinate-point selection during INR training (cf. our line-aware sampling).

Competing alternatives. TransformerPayne [Różański et al., 2025] demonstrates, for stellar spectral emulation specifically, that attention over the wavelength axis breaks the $\sim$1% plateau of MLP emulators with better accuracy *and* data efficiency, by capturing long-range correlations (co-varying lines of the same ion) that our independent per-line embeddings ignore – the best-motivated architecture competitor, modulo heavier inference. Yang et al. [2025] argue that for dense, regularly sampled signals, explicit grids plus interpolation beat implicit networks per unit compute – consistent with our interpolation baselines – which legitimises a grid/low-rank component plus small neural residual as the Tier-3 architecture, and strengthens the fixed-grid distillation route. Kolmogorov–Arnold networks [Liu et al., 2024b] report favourable scaling on smooth scientific fitting problems, but wall-clock efficiency claims replicate poorly and nothing suggests an advantage on line-forest targets (low priority). Attention/state-space operators [e.g. Wu et al., 2024] become relevant at the joint multi-element stage, not for 1-D single-element emulation.

Proposed tests. These candidates are *not* suited to a single factorial ablation: Schedule-Free *replaces* WSD+EMA (mutually exclusive alternatives, not additive switches), muP is testable only as a width sweep, and InfoBatch interacts with the sampler. The plan is instead: (1) A/B with 2–3 seeds at t04 scale for InfoBatch-style rescaling (success criterion: grating regression absent at full prioritization strength, equal MRE at lower cost); (2) a three-arm optimizer bake-off (WSD+EMA / Schedule-Free / Muon) at short budget, winner confirmed once at 100k steps; (3) a three-width muP sweep with the learning rate tuned only at the smallest width; (4) a FINER $\times$ curriculum 2×2 (FINER may subsume the coarse-to-fine schedule). Winners are composed greedily and confirmed in a single big_trunk-scale run, with the test set touched only at that final step.

11 Summary of discarded approaches

Reproducibility

Code: `spexai` branch `neural_operator`. Trainers: `spexai/train/train_operator.py` (variants/ablation), `train_adaptive.py` (prioritized sampling, gated augmentation, Tier 1/2 recipes), `train_broadened2.py` (revised (T, v) emulator). Benchmarks: `scripts/benchmark_operator.py` (native grid), `benchmark_broadening.py`, `benchmark_instruments.py` (resolution-matched + RMF-folded), `baseline_interpolation.py`. Sweep driver: `scripts/run_all_elements.py`. The recipe-screening program of Section 10 is implemented in `scripts/run_recipe_program.py` (14 arms; new options `--optimizer adamw|schedulefree|muon`, `--mup`, `--pr_correct`, `--activation finer` in `train_adaptive.py` and `spexai/train/optim.py`); confirmation on fresh off-grid PCHIP probes rather than the test set is `scripts/benchmark_offgrid.py`. Every training run writes train-vs-validation curves and three-temperature spectrum/residual/line-zoom figures automatically (`spexai/train/diagnostics.py`). The winning configuration (`big_trunk`): GELU 512×6 trunk, $K = 1024$ Fourier frequencies ($f_{\max} = 4000$), FiLM, trend head, per-bin

Approach	Evidence against	Replacement
Sobolev derivative loss	MRE 0.91% $\rightarrow$ 0.52% on removal	plain relative-error Huber
PCA(64)+GP baseline/generator	saturates at 0.2%; 9–17% at small n	per-bin PCHIP
Sparse-matrix broadening	2.8% MRE, 20% flux misplaced at $1000\,\mathrm{km\,s^{-1}}$	FFT / hybrid
Monolithic (T, v) emulator (v1, v2)	v1 8×, v2 (fixed) 4–6× worse than hybrid	hybrid (broaden outside the net)
Line-head enlargement	0.21% vs 0.16% (worse incl. line MRE)	trunk scaling (big_trunk)
Uniform batch sampling	0.45% vs 0.32% (reweight) at $n = 300$	error-prioritized sampling
Augmentation on dense grids	interp. error 100× below model error; train=val MRE	augmentation only for sparse grids
HPO's "big models diverge"	t16 stable and winning at lr 10^{-3}	decouple size from learning rate
20k-step cosine budgets	best checkpoint = final step, twice	100k+ steps; WSD schedule
Unweighted per-channel means	30× swings between models with equal medians	flux-weighted MRE (planned)
(T, v)-emulator line-head removal	core errors ±2–5% at all v	analytic Gaussian line head

Table 5: Approaches tried and discarded, with the deciding evidence.

normalisation, line head (dim 16, hidden 128), batch 128, 2048 points/spectrum, AdamW at 10^{-3} with WSD schedule, EMA decay 0.999, prioritized sampling with mix 0.4, 100k steps.

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